Lecture 19 - Reward model & Linear Dynamical System

Outline:

- State action models
- Finite horizon MDP
- Linear dynamical system
  - Model
  - LQR (Linear quadratic regularization)

State action rewards

$$R: S \times A \longmapsto \mathbb{R}$$

Bellman's Equation

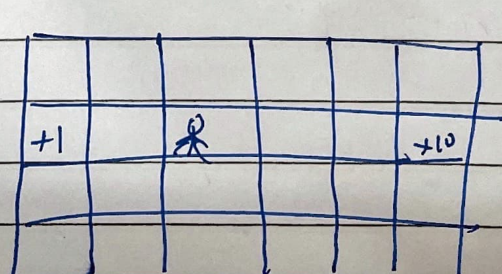
$$V^*(s) = \max_a \Big[\underbrace{R(s,a)}_{\text{immediate reward}} + \underbrace{\gamma \sum_{s'} P_{sa}(s') V^*(s')}_{\text{future rewards}} \Big]$$

$$\pi^*(s) = \arg\max_a R(s,a) + \gamma \sum_{s'} P_{sa}(s') V^*(s')$$

Finite Horizon MDP

$$(S, A, \{P_{sa}\}, T, R)$$

$$E\Big[R(s_0, a_0) + R(s_1, a_1 \ldots) + \ldots R(s_T, a_T)\Big]$$



where the robot will go depends on time left in the clock.

$\pi_t^*(s) \rightarrow$ optimal action depends on time
$\downarrow$
Its a non-stationary policy, it changes over time.

$\pi^*(s)$ - stationary policy, does not change over time.

Non stationary state transition

$$S_{t+1} \sim P^{(t)}_{s_t a_t}$$

$$R^{(t)}(s,a)$$

- Changing dynamics
- weather forecast
- Industrial automation

$$V_t^*(s) = E[\, R(s_t, a_t) + R(s_{t+1}, a_{t+1}) \ldots \; R(s_T, a_T) \mid \pi^*, s_0 = s \,]$$

Expected total payoff starting in state s at time t, execute π^*

Value iteration becomes a dynamic programming problem.

$$V_t^*(s) = \max_a R(s,a) + \sum_{s'} P_{sa}(s') \, V^*_{t+1}(s')$$

$$\pi^*(s) = \arg\max_a \left[R(s,a) + \sum_{s'} P_{sa}(s') \, V^*_{t+1}(s') \right]$$

$V_T^*(s) =$

at the final step, when clock is about to run out.

$$V_T^*(s) = \max_a R(s,a)$$

Linear Quadratic Regularization (LQR)

$(S, A, \{P_{sa}\}, T, R)$

$S = \mathbb{R}^n$ $\quad A = \mathbb{R}^d$

P_{sa}: $\quad s_{t+1} = \underbrace{A}_{\mathbb{R}^{n\times n}} s_t + \underbrace{B}_{\mathbb{R}^{n\times d}} a_t + \underbrace{w_t}_{w_t \sim \mathcal{N}(0, \Sigma_w)}$

$R(s,a) = -(s^T U s + a^T V a)$

where $U \in \mathbb{R}^{n\times n}$, $V \in \mathbb{R}^{d\times d}$

$U, V \geq 0$ → positive semi-definite

so, $s^T U s \geq 0$, $a^T V a \geq 0$

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want $s \simeq 0$

choose $U = I$, $V = I$

$$R(s,a) = -\left(\|s\|^2 + \|a\|^2\right)$$

Where to get A, B?

$$s_0^{(1)} \xrightarrow{a_0} s_1^{(1)} \xrightarrow{a_1} \cdots \quad s_T^{(1)}$$

$\vdots$

$$s_0^{(m)} \xrightarrow{a_0^{(m)}} \cdots$$

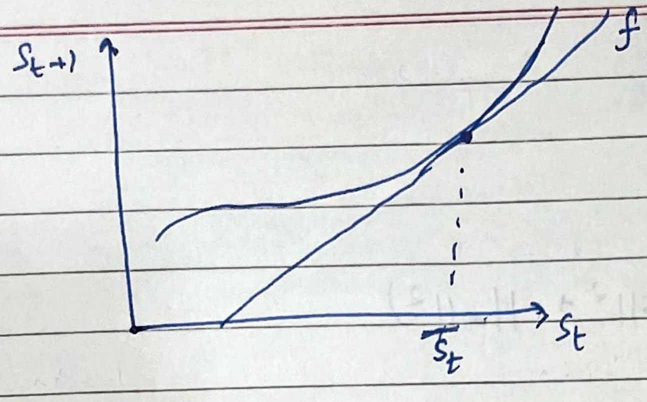
$$s_{t+1} \approx A s_t + B a_t$$

$$\min_{A,B} \sum_{i=1}^{m} \sum_{t=0}^{T-1} \left\| s_{t+1} - \left(A s_t + B a_t\right) \right\|^2$$

Linearize a non-linear model

$$s_{t+1} = f(s_t, a_t)$$

$$\begin{bmatrix} x_{t+1} \\ \dot{x}_{t+1} \\ \theta_{t+1} \\ \dot{\theta}_{t+1} \end{bmatrix} = f\left(\begin{bmatrix} x \\ \dot{x}_t \\ \theta_t \\ \dot{\theta}_t \end{bmatrix}, a_t \right)$$



$$s_{t+1} \simeq f'(\bar{s}_t) \cdot (s_t - \bar{s}_t) + f(\bar{s}_t)$$

$\bar{s}_t$ is a constant

Linear f around $(\bar{s}_t, \bar{a}_t)$

$$s_{t+1} \simeq f(\bar{s}_t, \bar{a}_t)$$
$$+ (\nabla_s f(\bar{s}_t, \bar{a}_t))^T (s_t - \bar{s}_t) +$$
$$(\nabla_a f(\bar{s}_t, \bar{a}_t))^T (a_t - \bar{a}_t)$$

~~s_{t+1}~~ $s_{t+1} = As_t + Ba_t$

$$s_t = \begin{pmatrix} x \\ \dot{x} \\ \theta \\ \vdots \\ \theta \end{pmatrix}$$

Problem:

$$S_{t+1} = As_t + Ba_t + \overset{\text{noise}}{w_T}$$

$$R(s,a) = -(s_t U_s + a^T V_a)$$

Total payoff: $R(s_0, a_0) \dots R(s_T, a_T)$

~~Dynam~~ Dynamic Programming →

$$V_T^*(s_T) = \max_{a_T} R(s_T, a_T)$$

$$= \max_{a_T} -s_T^T U s_T - \underbrace{a_T^T V a_T}_{\geq 0}$$

$$= -s_T^T U s_T$$

$$\pi_T^*(s_T) = \vec{0}$$

Suppose

$$\cancel{V^*_{t+1}} \quad V^*_{t+1}(s_{t+1}) = s_{t+1}^T \Phi_{t+1} s_{t+1} + \Psi_{t+1}$$

where $\Phi_{t+1} \in \mathbb{R}^{n\times n}$, $\Psi_{t+1} \in \mathbb{R}$

$$V_t^*(s_t) = \max_{a_t} R(s_t, a_t) + E_{s_{t+1} \sim P_{s_t a_t}} \left[V^*_{t+1}(s_{t+1}) \right]$$

$$V_t^*(s_t) = \max_{a_t} -s_t U s_t \cancel{+} - a_t V a_t + E_{s_{t+1} \sim \mathcal{N}(As_t + Ba_t, \Sigma)} \left[\underbrace{s_{t+1}^T \Phi_{t+1} s_{t+1} + \Psi_{t+1}}_{V^{*'}_{t+1}(s_{t+1})} \right]$$

$$s_{t+1} = As_t + Ba_t + w$$

$$w_t \sim \mathcal{N}(0, \Sigma_w)$$

The big quadratic function of a_t. Take derivatives wrt a_t. Set to 0, solve for a_t

$$a_t = \underbrace{\left(B^T \Phi_{t+1} B - V\right)^{-1} B^T \Phi_{t+1} A}_{L_t} \cdot s_t$$

$$\pi_t^*(s_t) = L_t \cdot s_t$$

$$V_t^*(s_t) = s_t^T \Phi_t s_t + \Psi_t$$

where

$$\Phi_t = A(\Phi_{t+1}, \Phi_{t+1}, B(B^T \Phi_t B - V)^{-1} B \Phi_{t+1}) A_t$$

$$\Psi_t = -tr\, \Sigma_w \Phi_{t+1} + \Psi_{t+1}$$

$\hookrightarrow$ Covariance of w_t

To summarize:-

initialize $\Phi_T = -u$ $\Psi_T = 0$

Recursive calculate

Φ_t, Ψ_t using Φ_{t+1}, Ψ_{t+1} (for $t = T-1, T-2, \ldots 0$)

Calculate L_t

$$\pi^*(s_t) = L_t \cdot s_t$$